JERRY LABS · AP BIOLOGY COMPLETE UNIT GUIDE

AP Biology Unit 4: Cell Communication & Cell Cycle

Signal Transduction · GPCRs & RTKs · Second Messengers · Feedback · Mitosis · Checkpoints & Cancer

AP Biology Unit 4 Study Guide

Cell Communication + Cell Cycle

Original study guide based on: OpenStax *Biology for AP® Courses*, companion readings Ch. 11 Cell Communication and Ch. 12 The Cell Cycle. This guide paraphrases and reorganizes the source material for review.

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1. Unit map: the big picture

Core storyline

1. **A signal exists**: a cell releases a ligand or makes direct contact.
2. **A target detects it**: only cells with the matching receptor respond.
3. **The message is relayed**: receptor activation changes intracellular proteins, second messengers, and phosphorylation states.
4. **The cell responds**: gene expression, metabolism, movement, secretion, growth, division, or apoptosis changes.
5. **The response stops**: ligands are removed, second messengers are degraded, and proteins are dephosphorylated.
6. **Cell-cycle control uses the same logic**: external growth signals and internal checkpoints determine whether a cell grows, copies DNA, and divides.

Unit map

Anchor idea	Must connect to	Typical AP evidence move
Reception	Ligand + specific receptor	Predict which cells respond
Transduction	Relay proteins, kinases, second messengers	Trace a mutation or inhibitor downstream
Response	Protein activity or gene expression	Explain a phenotype from pathway data
Termination	Degradation, reuptake, phosphatases, pumps	Predict effects of a stuck-on signal
Cell cycle	G1, S, G2, M, cytokinesis	Track DNA content and chromosome structure
Checkpoints	Cyclins, Cdks, p53, p21, Rb	Explain arrest, apoptosis, or cancer
Division	Mitosis or binary fission	Compare chromosome segregation mechanisms

AP Biology emphasis: Build explanations from a model. When given an unfamiliar pathway, identify the receptor, the changed component, the direction of the change, and the final cellular response.

2. Cell communication: distance and delivery

Signaling mode	Distance	Signal route	Example / key use
Direct contact	Adjacent cells	Gap junctions in animals; plasmodesmata in plants	Small molecules and ions coordinate neighbors
Autocrine	Same cell or nearby same-type cells	Secreted ligand returns to matching receptors	Self-amplification; immune and developmental signaling
Paracrine	Short range	Diffusion through extracellular space or synapse	Neurotransmitters; local immune signals
Endocrine	Long range	Hormones travel through blood	Coordinated response across organs

Plain-language translation

- **Autocrine = “me talking to me.”**
- **Paracrine = “talking to nearby cells.”**
- **Endocrine = “broadcasting through circulation.”**
- **Direct contact = “sharing a channel.”**

Distance does not alone determine response speed. Synaptic signaling is short-range and often rapid because the synaptic gap is tiny. Endocrine signaling is usually slower to arrive but can last longer. Signal removal, receptor type, and target-cell machinery also matter.

Direct-contact diagram

```text
Animal cell A Animal cell B
[cytoplasm]--gap junction----[cytoplasm]
small ions / mediators move directly
Plant cell A Plant cell B
[cytoplasm]--plasmodesma----[cytoplasm]
```

Only small molecules fit through these channels; large proteins and DNA do not pass through them.

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3. Reception: ligand, receptor, and cell specificity

A **ligand** is a signaling molecule that binds a receptor. A **receptor** is a protein whose shape and chemistry allow a particular signal to be recognized. A target cell may ignore a ligand simply because it lacks the correct receptor or because its downstream proteins differ.

Internal versus cell-surface receptors

Feature	Internal receptor	Cell-surface receptor
Ligand tendency	Small, hydrophobic, membrane-permeable	Polar, charged, or large and membrane-impermeable
Binding location	Cytoplasm or nucleus	Extracellular domain of membrane protein
Common immediate effect	Ligand–receptor complex regulates transcription	Conformational change activates intracellular relay
Typical timing	Often slower onset, longer-lasting gene-expression effects	Can be fast or slow depending on pathway
Examples	Steroid hormones, thyroid hormone, vitamin D	Neurotransmitters, growth factors, many peptide hormones

Important nuance: “Hydrophobic ligand” predicts membrane crossing, but it does not guarantee one specific response. The cell’s receptor and gene-expression machinery still determine the outcome.

Three major surface-receptor classes

1. **Ion-channel-linked receptors:** ligand binding opens or closes a pore; ions move down electrochemical gradients.
2. **G-protein-linked receptors (GPCRs):** ligand binding activates a G protein; GDP is exchanged for GTP; downstream enzymes or channels change.
3. **Enzyme-linked receptors:** ligand binding often brings receptors together; intracellular kinase domains or associated enzymes phosphorylate targets. Receptor tyrosine kinases (RTKs) are a major

example.

4. Signal transduction: the relay system

Signal transduction converts receptor activation into intracellular change. A typical pathway includes:

```text
ligand -> receptor -> relay protein -> second messenger / kinase cascade
-> target protein or transcription factor -> cellular response
-> signal termination
```

High-yield vocabulary

Term	Meaning in plain language
Conformational change	A protein changes shape, altering its activity or binding partners
Phosphorylation	A kinase adds a phosphate group; the target may activate or inactivate
Dephosphorylation	A phosphatase removes a phosphate group
Second messenger	Small intracellular signal such as cAMP, Ca2+, IP3, or DAG
Amplification	One activated component activates many copies downstream
Integration	Inputs from multiple pathways combine before a response
Branching	One pathway influences multiple outputs
Upstream	Earlier in the pathway
Downstream	Later in the pathway

G-protein cycle

1. Ligand binds GPCR.
2. Receptor changes shape.
3. G protein's alpha subunit exchanges GDP for GTP.
4. Alpha and/or beta-gamma subunits activate an effector.
5. Effector changes an ion channel or enzyme.
6. GTP is hydrolyzed to GDP; the signal ends and subunits reassociate.

AP reasoning shortcut: If GTP hydrolysis is blocked, the active G protein may remain active too long, producing an exaggerated or prolonged response.

5. Second messengers and phosphorylation

cAMP route

```text
receptor -> adenylyl cyclase -> ATP -> cAMP -> protein kinase A
-> phosphorylation of target proteins -> response
-> phosphodiesterase converts cAMP to AMP
```

IP3/DAG/Ca2+ route

```text
activated receptor -> phospholipase C -> PIP2 split
/ \
IP3 DAG
ER Ca2+ channel opens activates PKC
Ca2+ rise -> target proteins
```

- **cAMP** is made from ATP and activates protein kinase A.
- **Ca2+** is normally kept low in the cytoplasm; a sudden increase can trigger secretion or contraction.
- **IP3** diffuses through the cytoplasm and can release Ca2+ from the endoplasmic reticulum.
- **DAG** remains in the membrane and helps activate protein kinase C.

The same second messenger can produce different results in different cell types because the target proteins and genes differ.

Termination checklist

- Degrade or remove the ligand.
- Reuptake neurotransmitters or digest them enzymatically.
- Hydrolyze GTP to GDP.
- Dephosphorylate relay proteins.
- Convert cAMP to AMP with phosphodiesterase.
- Pump Ca2+ back into storage or out of the cell.

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6. Cellular responses and pathway disruptions

A signaling pathway can change:

- **Gene expression:** transcription factors enter or act in the nucleus.
- **Metabolism:** epinephrine/cAMP signaling promotes glycogen breakdown for rapid glucose availability.

- **Growth and division:** growth factors activate RTKs and downstream MAP kinase pathways.
- **Secretion or contraction:** Ca2+ can trigger insulin release or muscle contraction.
- **Apoptosis:** cells with severe damage or inappropriate developmental identity can be removed.

Disease logic examples

Disruption	Immediate pathway effect	Likely outcome
Cholera toxin modifies a G protein	Chloride secretion remains active	Severe water and ion loss
RAS cannot hydrolyze GTP	RAS remains active	Persistent downstream proliferation signals
HER2/RTK is overactive	Growth pathway is repeatedly stimulated	Excessive cell division
Receptor is blocked	Reception fails	Downstream response decreases or disappears
Phosphatase is inhibited	Phosphorylated proteins persist	Signal may be prolonged
p53 is defective	DNA damage checkpoint weakens	Damaged cells continue dividing

Causal-chain template: “Because component X is unable to [activate/inactivate/terminate], component Y remains [active/inactive]. Therefore, the cell [does/does not] perform response Z.”

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7. Cell-cycle map and chromosome bookkeeping

Cell-cycle sequence

```text
G1 --growth, organelles, resources--> S --DNA replication--> G2
^ | |
| v v
G0 <----exit if not dividing sister chromatids Mitosis
attached + cytokinesis
```

Phase	What the cell is doing	Chromosome note
G1	Growing; making proteins and resources	Each chromosome has one chromatid
S	Replicating nuclear DNA; centrosomes begin duplicating	Each chromosome now has two sister chromatids
G2	Checking replication; making division proteins; final growth	Two chromatids per chromosome remain joined
Mitosis	Separating sister chromatids into two nuclei	After separation, each chromatid is a chromosome
Cytokinesis	Dividing cytoplasm	Two daughter cells form
G0	Quiescent state outside active cycle	May be temporary or long-term

Ploidy versus DNA amount

- **Ploidy** counts chromosome sets and generally stays constant through mitosis.
- **DNA amount** doubles during S phase because each chromosome is copied.
- A replicated chromosome is still counted as **one chromosome** until sister chromatids separate.
- Sister chromatids are identical copies; homologous chromosomes are maternal/paternal partners with the same loci but potentially different alleles.

8. Mitosis and cytokinesis process table

Stage	Defining events	What to look for
Prophase	Chromatin condenses; centrosomes move apart; spindle begins	Visible duplicated chromosomes
Prometaphase	Nuclear envelope fragments; kinetochores attach spindle microtubules	Spindle attachment
Metaphase	Chromosomes align at metaphase plate; sister chromatids still joined	“Middle” alignment
Anaphase	Cohesin breaks down; sister chromatids separate and move poleward	Two chromosome groups moving apart
Telophase	Chromosomes arrive, decondense; nuclear envelopes reform	Two forming nuclei
Cytokinesis	Cytoplasm divides	Furrow in animals; cell plate in plants

Animal versus plant cytokinesis

Animal cell	Plant cell
Actin–myosin contractile ring forms	Golgi vesicles gather at the center
Cleavage furrow pinches inward	Phragmoplast guides vesicles
Two membranes separate	Cell plate grows outward and becomes new wall/membranes

Binary fission comparison

Prokaryotes generally have one circular chromosome in a nucleoid rather than multiple nucleus-enclosed chromosomes. DNA replication and cell elongation separate chromosome copies; an **FtsZ ring** directs division. There is no karyokinesis because there is no nucleus to divide. FtsZ and eukaryotic tubulin share structural and functional ancestry.

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9. Checkpoints and cell-cycle control

Checkpoint	Main question	If the answer is “no”
G1 / restriction point	Is the cell large enough, resourced, and free of major DNA damage?	Repair, pause, or enter G0
G2/M	Has DNA been completely and accurately replicated?	Repair or arrest before mitosis
M / spindle checkpoint	Are all kinetochores attached to opposite spindle poles?	Delay anaphase

Positive and negative regulation

- **Cyclins** rise and fall during the cycle.
- **Cdks** are kinases whose activity depends on binding the correct cyclin and receiving activating phosphorylation.
- **Cdk inhibitors** prevent premature progression.
- **p53** responds to DNA damage: pause for repair or trigger apoptosis if damage is severe.
- **p21** inhibits cyclin–Cdk complexes downstream of p53.
- **Rb** restrains E2F when active; phosphorylation releases E2F so genes needed for G1/S progression can be expressed.

Control diagram

```text
DNA damage -> p53 rises -> p21 rises -> cyclin-Cdk inhibited -> G1 arrest
\-> repair succeeds -> cycle resumes
\-> repair fails -> apoptosis
```

```

A checkpoint is not a single “sensor” that guarantees perfection. It is a control point where multiple conditions are evaluated before an irreversible transition.

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## 10. Cancer as failed regulation

Cancer is not caused by one universal pathway error. It results from accumulated changes that allow cells to evade normal controls, divide excessively, survive when they should die, or invade other tissues.

- A **proto-oncogene** normally promotes cell growth or division. A gain-of-function mutation can convert it into an **oncogene**.
- A **tumor-suppressor gene** normally restrains division, repairs damage, or promotes apoptosis. Loss of function removes a brake.
- Mutant **p53** may fail to detect damage, recruit repair, or trigger apoptosis.
- Excessive RTK/HER2 signaling or permanently active RAS can push growth pathways forward.

### Common causal comparison

| Normal condition | Cancer-associated change |
|---|---|
| Growth factor is present only when needed | Growth receptor or downstream protein is active without the signal |
| DNA damage pauses the cycle | p53/p21 pathway fails |
| Spindle attachment is checked | Cells proceed with faulty chromosome segregation |
| Unneeded cells undergo apoptosis | Apoptosis is avoided |
| Division is limited by checkpoints | Checkpoint controls are bypassed |

### What an AP explanation should include

1. Name the altered regulator or pathway component.
2. State whether its activity increases or decreases.
3. Identify the checkpoint or response affected.
4. Connect that change to continued division, mutation accumulation, or survival.

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## 11. Misconceptions to eliminate

1. **“Interphase is a resting phase.”** False. G1, S, and G2 include growth, biosynthesis, DNA replication, and preparation.
2. **“A replicated chromosome is two chromosomes.”** Not until sister chromatids separate in anaphase.
3. **“Mitosis always produces a new organism.”** Mitosis supports growth, repair, replacement, and sometimes asexual reproduction.
```

4. **“Homologous chromosomes are identical copies.”** They carry the same genes at corresponding loci but may have different alleles.
5. **“Ploidy doubles during S phase.”** DNA amount doubles; chromosome-set number does not.
6. **“All ligands enter cells.”** Polar or large ligands usually bind surface receptors.
7. **“A receptor is the same as a second messenger.”** The receptor detects the external signal; the second messenger relays information inside.
8. **“Phosphorylation always activates a protein.”** It can activate or inhibit, depending on the target and context.
9. **“One ligand has one effect everywhere.”** Different target cells express different receptors and downstream proteins.
10. **“Signal transduction always amplifies.”** Cascades often amplify, but pathways can also integrate, branch, dampen, or terminate signals.
11. **“Apoptosis is accidental cell death.”** Apoptosis is regulated, programmed cell death.
12. **“Cancer is just fast growth.”** It also involves failed checkpoints, altered apoptosis, mutation accumulation, and sometimes invasion.
13. **“Plant cytokinesis uses a cleavage furrow.”** Plants build a cell plate because the cell wall prevents pinching inward.
14. **“Binary fission is mitosis without a nucleus.”** It is a distinct prokaryotic division process.
15. **“A checkpoint repairs every problem.”** It can pause, allow repair, send the cell to G0, or trigger apoptosis.

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12. Original multiple-choice questions

Choose the best answer.

1. A hormone is present in the bloodstream, but only liver cells increase glycogen breakdown. Which explanation is best?

A. Liver cells contain a receptor and relevant downstream proteins.

B. Liver cells contain more DNA than other cells.

C. The hormone enters only liver cells by diffusion.

D. The hormone is converted into a neurotransmitter in the liver.

1. A mutation prevents a GPCR-associated alpha subunit from hydrolyzing GTP. What is most likely?

A. The receptor cannot bind ligand.

B. The G protein remains active longer than normal.

C. cAMP is immediately converted to AMP.

D. The ligand becomes hydrophobic.

1. Which change most directly lowers a cAMP signal after receptor activation?

A. Inhibiting phosphodiesterase

B. Increasing adenylyl cyclase activity

C. Increasing phosphodiesterase activity

D. Opening an IP3-gated calcium channel

1. A steroid hormone binds an intracellular receptor. Which event is most likely downstream?

A. Direct opening of a sodium channel by the ligand

B. Ligand–receptor complex regulation of transcription

C. GDP exchange on a membrane G protein only

D. Cleavage of PIP2 at the membrane

1. A drug blocks RTK autophosphorylation but not ligand binding. Which result is most likely?

A. Downstream kinase recruitment decreases.

B. The ligand becomes unable to circulate.

C. The receptor moves into the nucleus and replicates DNA.

D. Cytokinesis immediately begins.

1. In a cell, IP3 is produced but cannot bind its ER receptor. Which immediate change is least likely?

A. Release of stored Ca2+ into cytoplasm

B. Activation of Ca2+-dependent proteins

C. A downstream response that depends on Ca2+

D. Formation of IP3 from PIP2

1. During S phase, a human somatic cell changes from 2n, 1x DNA to:

A. 1n, 1x DNA

B. 2n, 2x DNA with sister chromatids

C. 4n, 2x DNA with homologs paired

D. 2n, 1x DNA with no chromosomes

1. Which observation identifies metaphase?

A. Nuclear envelopes have reformed.

B. Sister chromatids have separated.

C. Duplicated chromosomes align at the cell equator.

D. DNA is being replicated.

1. A plant cell completes mitosis but cannot deliver Golgi vesicles to the center. Which process is impaired?

A. Formation of a cell plate

B. Formation of kinetochores

C. Separation of sister chromatids

D. DNA replication

1. A mutation disables the M checkpoint. Which event could occur prematurely?

A. Anaphase before all kinetochores attach correctly

B. DNA replication before G1

C. Cytokinesis before chromosome condensation in every case

D. G0 entry after every S phase

1. A cell has high p53 and p21 after DNA damage. What is the best prediction?

A. Cyclin-Cdk activity is inhibited and the cycle pauses.

B. RAS hydrolyzes GTP more slowly.

C. The cell immediately enters anaphase.

D. cAMP permanently increases.

1. Which change is most consistent with a proto-oncogene becoming an oncogene?

A. A growth-promoting protein becomes active without its normal signal.

B. A tumor suppressor becomes more effective at arresting the cycle.

C. A phosphatase turns off a signal more rapidly.

D. A damaged cell enters apoptosis sooner.

1. Two tissues respond differently to the same second messenger because they:

A. use different genetic material

B. contain different target proteins and regulatory networks

C. have different numbers of chromosome sets

D. prevent all phosphorylation

1. Which comparison between binary fission and mitosis is correct?

A. Both require a nucleus.

B. Binary fission uses a circular chromosome and no karyokinesis.

C. Mitosis uses an FtsZ ring instead of a spindle.

D. Binary fission produces four daughter cells.

1. A phosphatase inhibitor is added to a cell. Which outcome is most plausible?

A. Phosphorylated pathway components persist longer.

B. All ligands are unable to bind receptors.

C. DNA cannot be packaged into nucleosomes.

D. The cell cannot perform cytokinesis under any condition.

MCQ answer key with brief reasoning

1. **A** — receptor expression and downstream machinery determine target-cell response.
2. **B** — GTP hydrolysis normally terminates the active state.
3. **C** — phosphodiesterase removes cAMP.
4. **B** — intracellular receptors commonly regulate gene expression.

5. **A** — phosphorylation creates or activates downstream docking/relay functions.
6. **A** — IP3 must open ER channels to release stored Ca2+.
7. **B** — ploidy remains 2n while DNA amount doubles.
8. **C** — metaphase is equatorial alignment.
9. **A** — plant cytokinesis depends on vesicle fusion into a cell plate.
10. **A** — the spindle checkpoint delays anaphase until attachment is correct.
11. **A** — p21 inhibits cyclin-Cdk complexes.
12. **A** — constitutive growth signaling is a gain-of-function pattern.
13. **B** — target proteins and cell context differ.
14. **B** — prokaryotes divide their nucleoid-associated chromosome without nuclear division.
15. **A** — dephosphorylation is slowed.

13. Original free-response questions with answers

FRQ 1: Receptor pathway analysis

A researcher exposes two cell types to ligand L. Cell type X rapidly increases cytoplasmic Ca2+ and secretes a protein. Cell type Y shows no measurable change. A mutation is then introduced into cell type X that prevents phospholipase C from cleaving PIP2.

(a) Explain why X responds but Y does not.

(b) Describe the pathway from receptor activation to Ca2+ release in X.

(c) Predict the effect of the phospholipase C mutation on secretion and justify the prediction.

(d) Identify one mechanism that could terminate the Ca2+ signal.

Answer:

(a) X expresses the receptor for L and the downstream proteins needed for the response; Y may lack the receptor or a required relay component.

(b) L binds a surface receptor, activating phospholipase C. Phospholipase C cleaves PIP2 into IP3 and DAG. IP3 binds a channel on the ER, releasing Ca2+ into the cytoplasm. Ca2+ activates proteins that promote secretion; DAG can activate protein kinase C.

(c) Secretion would decrease or fail because PIP2 would not be cleaved, so IP3 would not form and ER Ca2+ release would be reduced. The receptor may still bind L, but the pathway is blocked downstream.

(d) Ca2+ pumps can return Ca2+ to the ER or export it; additional termination could include phosphatases, ligand removal, and degradation of second messengers.

FRQ 2: Cell-cycle control and cancer

A cultured cell line contains a mutation that makes p53 unable to activate p21. The cells are exposed to a DNA-damaging treatment. Compared with control cells, the mutant cells continue entering S phase at a high frequency.

(a) Explain why loss of p21 activation changes progression through the G1 checkpoint.

(b) Predict one effect on the genetic composition of later daughter cells.

(c) Explain how this mutation could contribute to tumor formation.

(d) A second treatment activates apoptosis only in cells with severe DNA damage. Explain why this treatment could reduce the mutant population.

Answer:

(a) p53 normally responds to DNA damage and increases p21. p21 inhibits cyclin-Cdk complexes, helping pause the cycle. Without p21 activation, cyclin-Cdk activity may remain high, allowing the cell to pass G1/S despite damage.

(b) Daughter cells may inherit unrepaired mutations or chromosome abnormalities because damaged DNA was replicated and segregated.

(c) Continued division propagates mutations. Additional changes may activate growth-promoting genes or disable other checkpoint and apoptosis mechanisms, allowing uncontrolled proliferation.

(d) The treatment would selectively remove cells carrying severe damage, reducing survival of cells that bypassed the p53-p21 arrest. This is an inference from the treatment's stated selectivity.

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14. Cram sheet: last-minute review

Signaling in one line

Reception → transduction → response → termination.

Receptor matching

- Hydrophobic ligand → often intracellular receptor → transcriptional effect.
- Polar/large ligand → usually surface receptor → relay pathway.
- Ion channel receptor → ion flow.
- GPCR → GDP/GTP switch; often cAMP, IP3, DAG, or Ca2+.
- RTK → dimerization/autophosphorylation; growth pathways.

Second messengers

- **cAMP:** adenylyl cyclase → PKA.
- **IP3:** ER Ca2+ release.
- **DAG:** membrane-associated PKC activation.
- **Ca2+:** secretion, contraction, enzyme activation.

Cell cycle

- **G1:** grow and gather resources.
- **S:** copy DNA; sister chromatids form.
- **G2:** verify/prepare.
- **M:** separate chromosomes.
- **Cytokinesis:** divide cytoplasm.

- **G0:** not actively preparing to divide.

Mitosis order

P–P–M–A–T = prophase, prometaphase, metaphase, anaphase, telophase.

Checkpoints

- G1: size, resources, DNA damage, growth signals.
- G2: DNA fully replicated and undamaged.
- M: spindle attachment at kinetochores.

Regulators

- Cyclin + Cdk = positive progression signal.
- p53 → p21 → Cdk inhibition.
- Rb active → E2F restrained.
- Tumor suppressor loss = brakes fail.
- Oncogene activation = accelerator stuck.

AP explanation sentence frame

"Because **[mutation/treatment]** changes **[component]**, **[downstream component]** remains/instead becomes **[active/inactive]**, so the cell's **[response]** increases/decreases. This is supported by **[data or model feature]**."

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